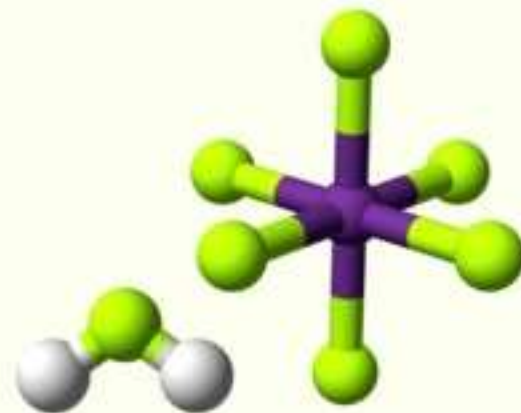


CHEMISTRY

CHEMICAL REACTIONS

MODULE - 13



CHANGES AROUND US



MELTING



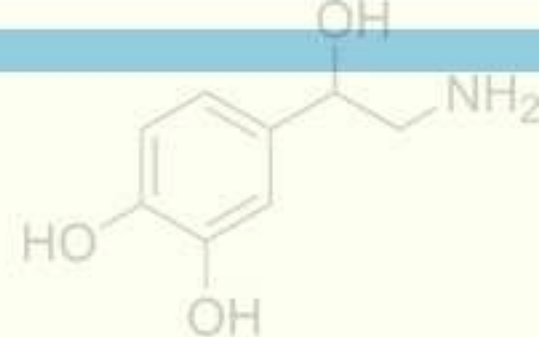
RUSTING



BURNING



GROWING



CHANGES AROUND US

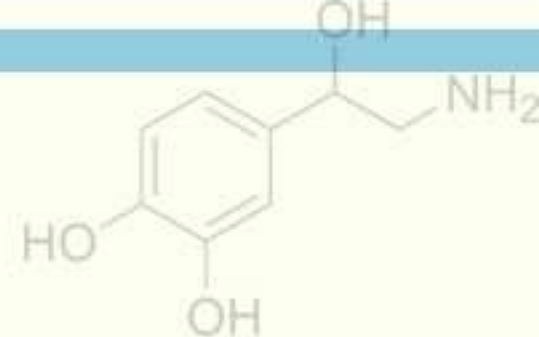
PHYSICAL CHANGE



CHEMICAL CHANGE



CONTRASTING FEATURES

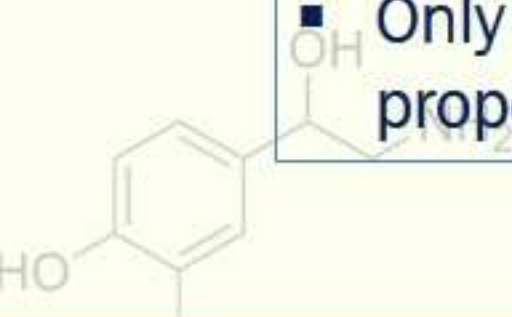


PHYSICAL CHANGE

- Physical changes are easy to reverse
- Produce no new substances
- Physical change produce no energy
- Only affect physical properties

CHEMICAL CHANGE

- Chemical changes are usually irreversible.
- Always produce new substances.
- Chemical change produces heat/light/sound energy
- Affect both physical and chemical property.



CHEMICAL CHANGE

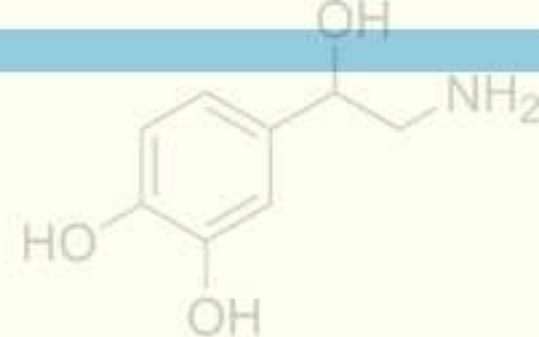
CHEMICAL REACTIONS



- Two or more substances react with each other to form new substances with different properties.
- There will be a change in
 - Colour
 - Odour
 - State
 - Temperature



HOW TO WRITE A CHEMICAL REACTION?



LEFT SIDE

RIGHT SIDE

C

3

3



H

8

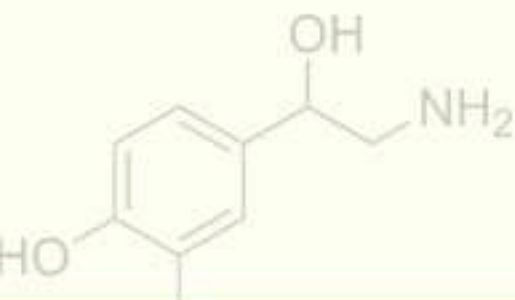
8



O

10

10



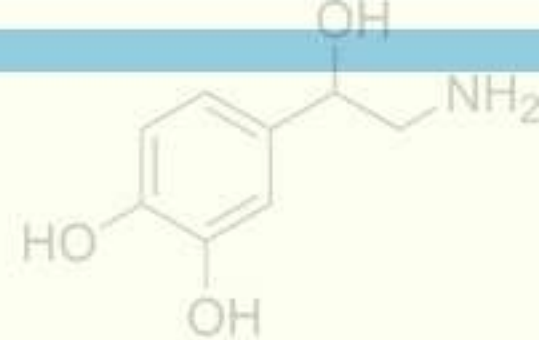
1 A chemical reaction can be written as word equation or chemical equation

2 Chemical equations are symbolic and are preferred.

3 It has reactants on its LHS and products on its RHS.

4 No. of atoms of each element on RHS should equal LHS.





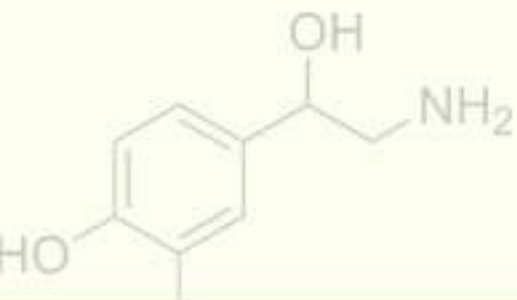
CHEMICAL REACTIONS

COMBINATION

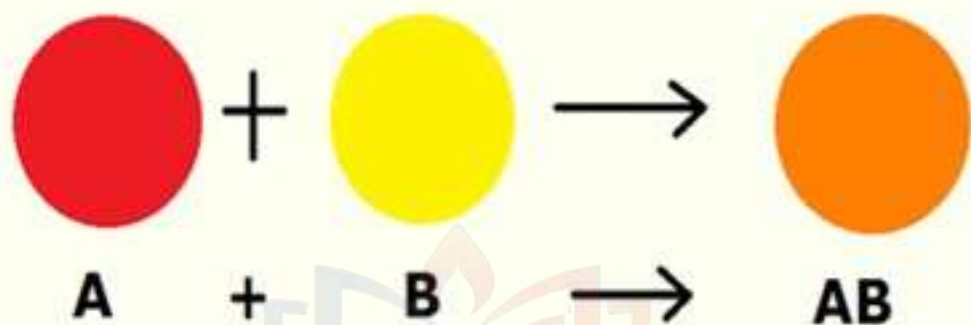
SINGLE
DISPLACEMENT

DECOMPOSITION

DOUBLE DISPLACEMENT



COMBINATION REACTION

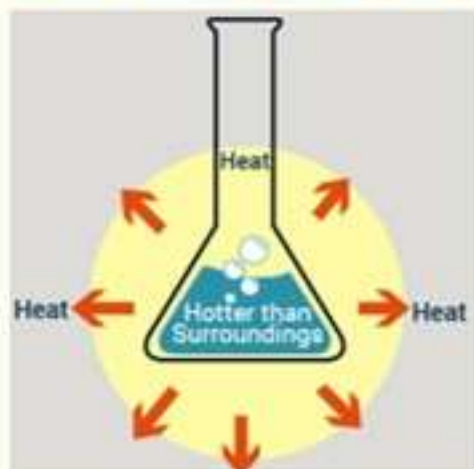


Combination reactions release energy and thus they are also exothermic reaction.

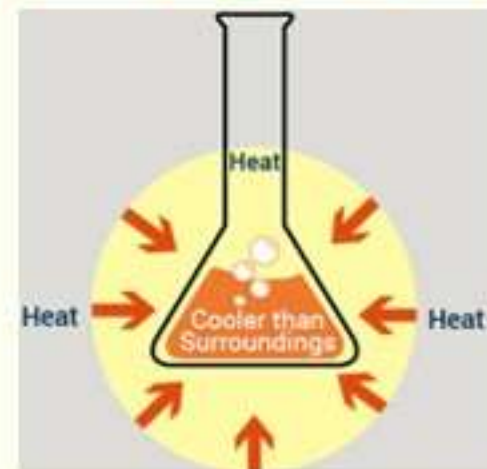
- The Reactants will combine to form products.
- This type of reactions are seen in
 - Combustion reaction
 - Oxidation reaction
 - Halogen addition
 - Photosynthesis



EXOTHERMIC AND ENDOTHERMIC REACTIONS



- Heat is evolved
- Hotter than the surroundings
- Products have less energy than reactants



- Heat is absorbed
- Cooler than the surroundings
- Products have more energy than reactants



DECOMPOSITION REACTIONS



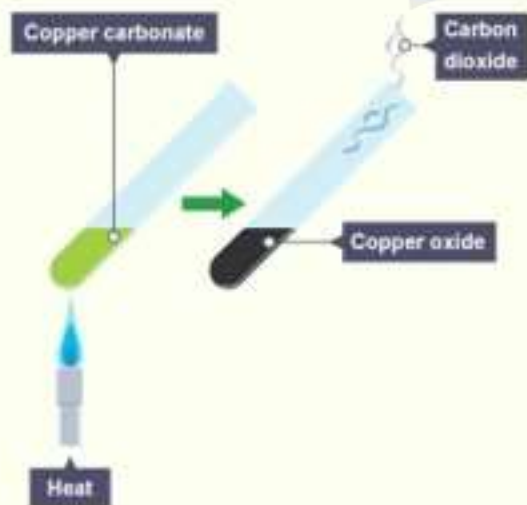
Decomposition reactions require energy to be supplied and thus it is called endothermic reaction.

- A reaction where a complex molecule break down to two or more simpler products.
- This process can be a single step or maybe multiple steps.
- Decomposition of inorganic compounds is a single step,
- Decomposition of food into energy in our body is a multi step process.

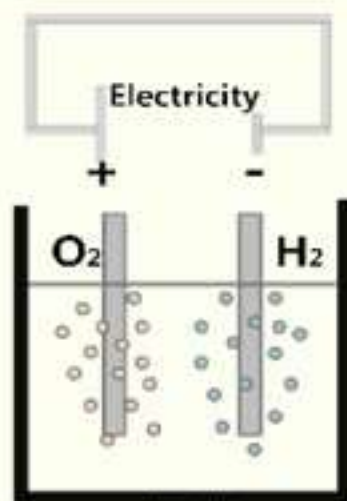


DECOMPOSITION REACTIONS

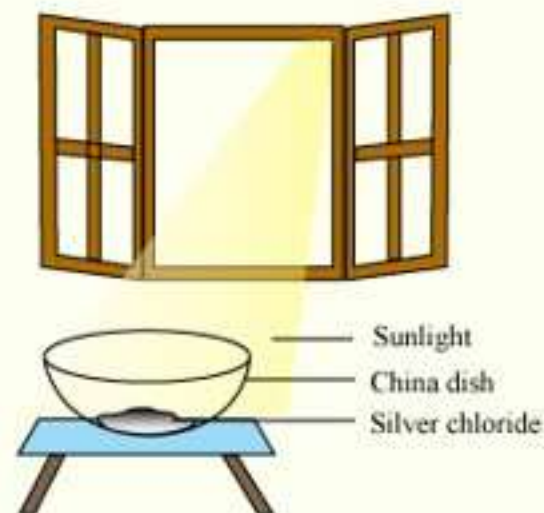
THERMAL



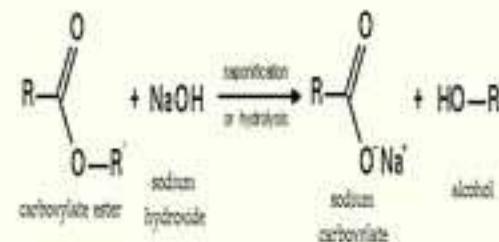
ELECTROLYTIC



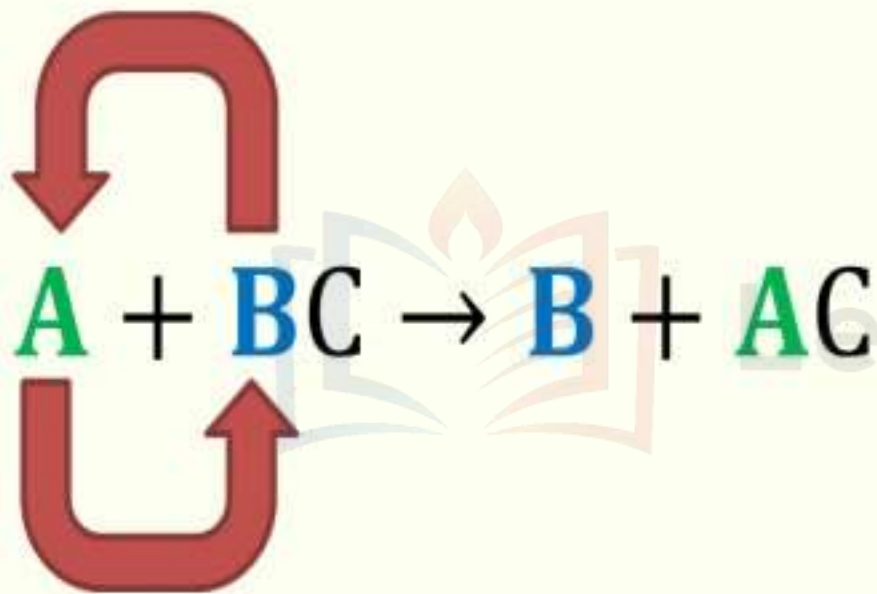
PHOTOLYTIC



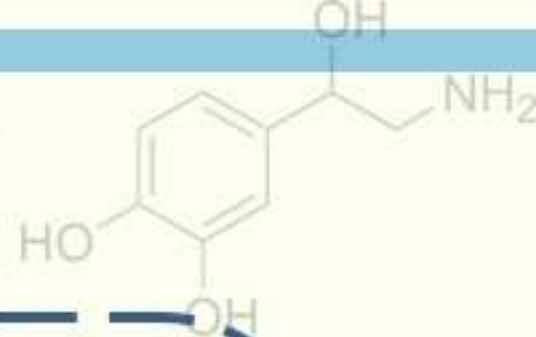
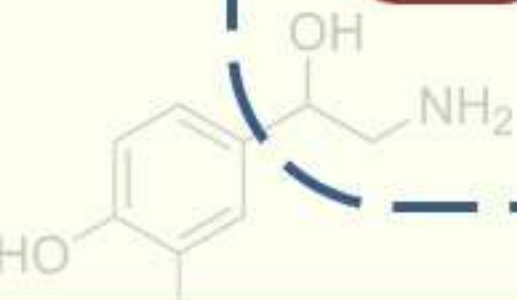
ACIDIC



SINGLE DISPLACEMENT REACTIONS

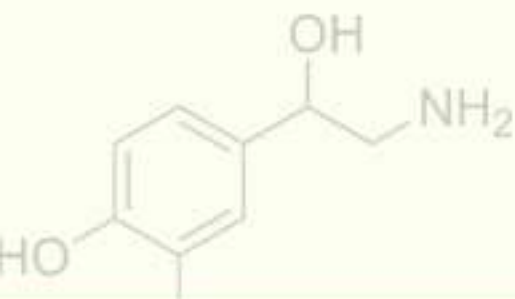
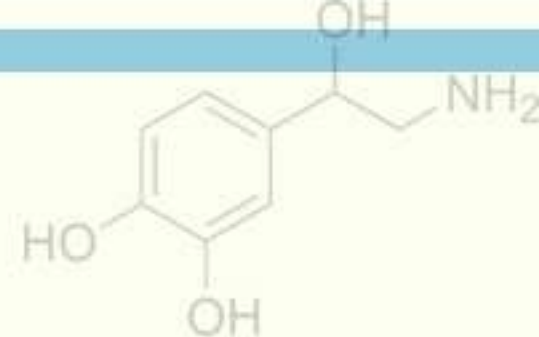


- The reaction of replacement of cations.
- A more stronger cation will replace or displace a less stronger cation.
- If the cation in solution is more strong, no reaction will occur.
- In some cases, displacement occur in anions too, such as halogens.

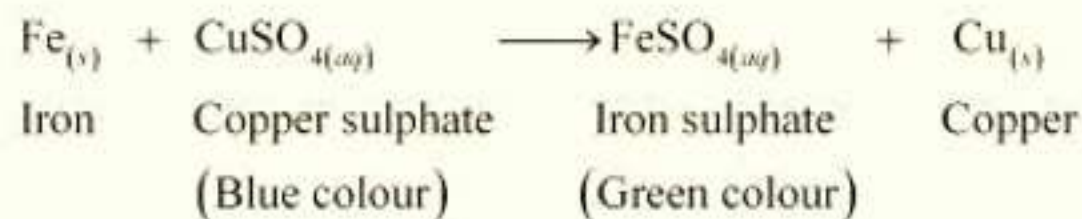
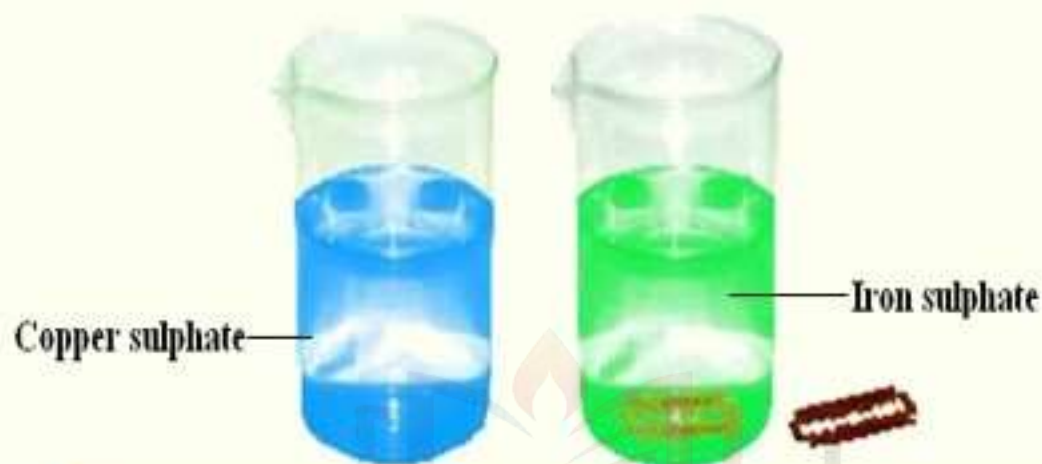
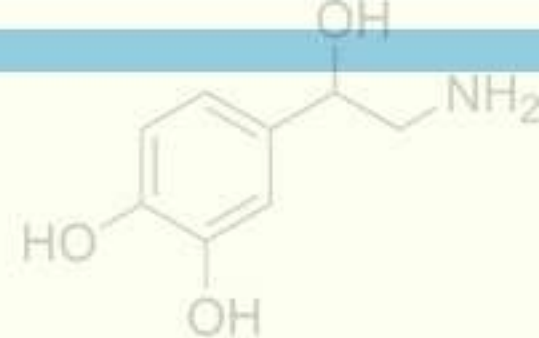


METAL REACTIVITY SERIES

potassium	most reactive	K	Kay
sodium		Na	Stop
calcium		Ca	Calling
magnesium		Mg	Me
aluminium		Al	A
carbon		C	Cute
zinc		Zn	Zebra,
iron		Fe	Forever
tin		Sn	So
lead		Pb	Put
hydrogen		H	Here(some)
copper		Cu	Copper
silver		Ag	Silver
gold		Au	Gold
platinum	least reactive	Pt	Platinum



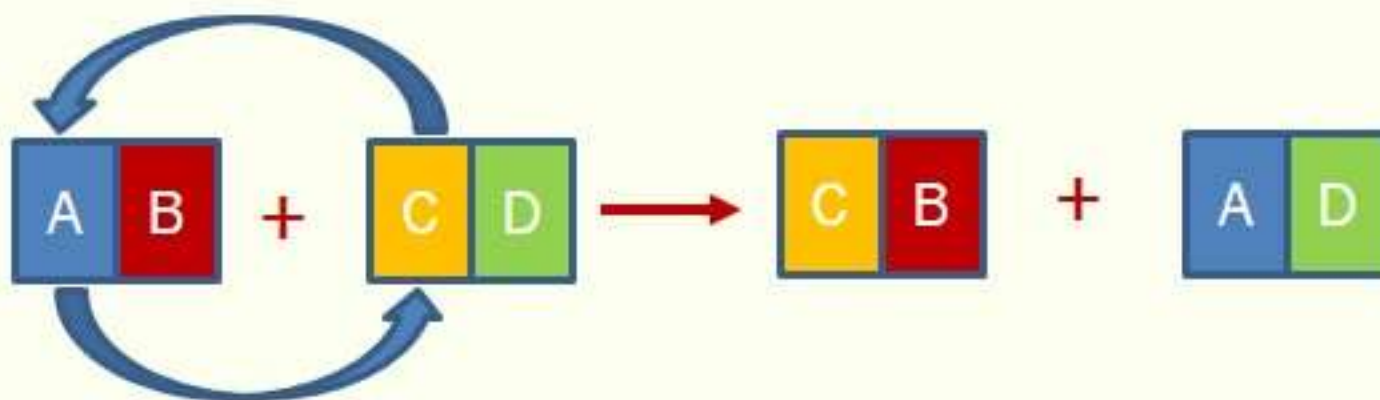
REACTION BETWEEN COPPER SULPHATE AND IRON



- Copper is less reactive than most metals.
- Copper can be displaced from its solution.
- But copper cannot displace any metal
- The displaced copper gets deposited as it comes out of the solution.
- The colour of the solution changes as the copper ions are displaced by iron ions.



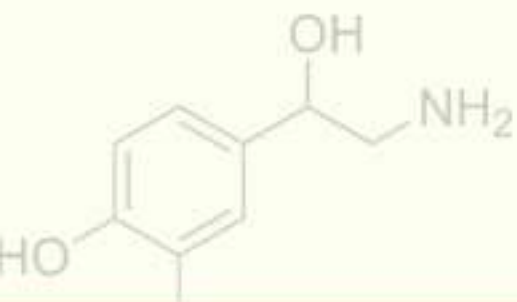
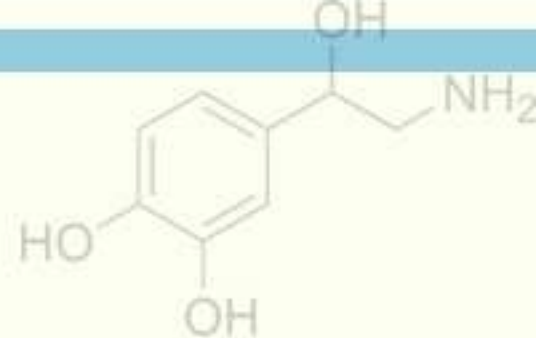
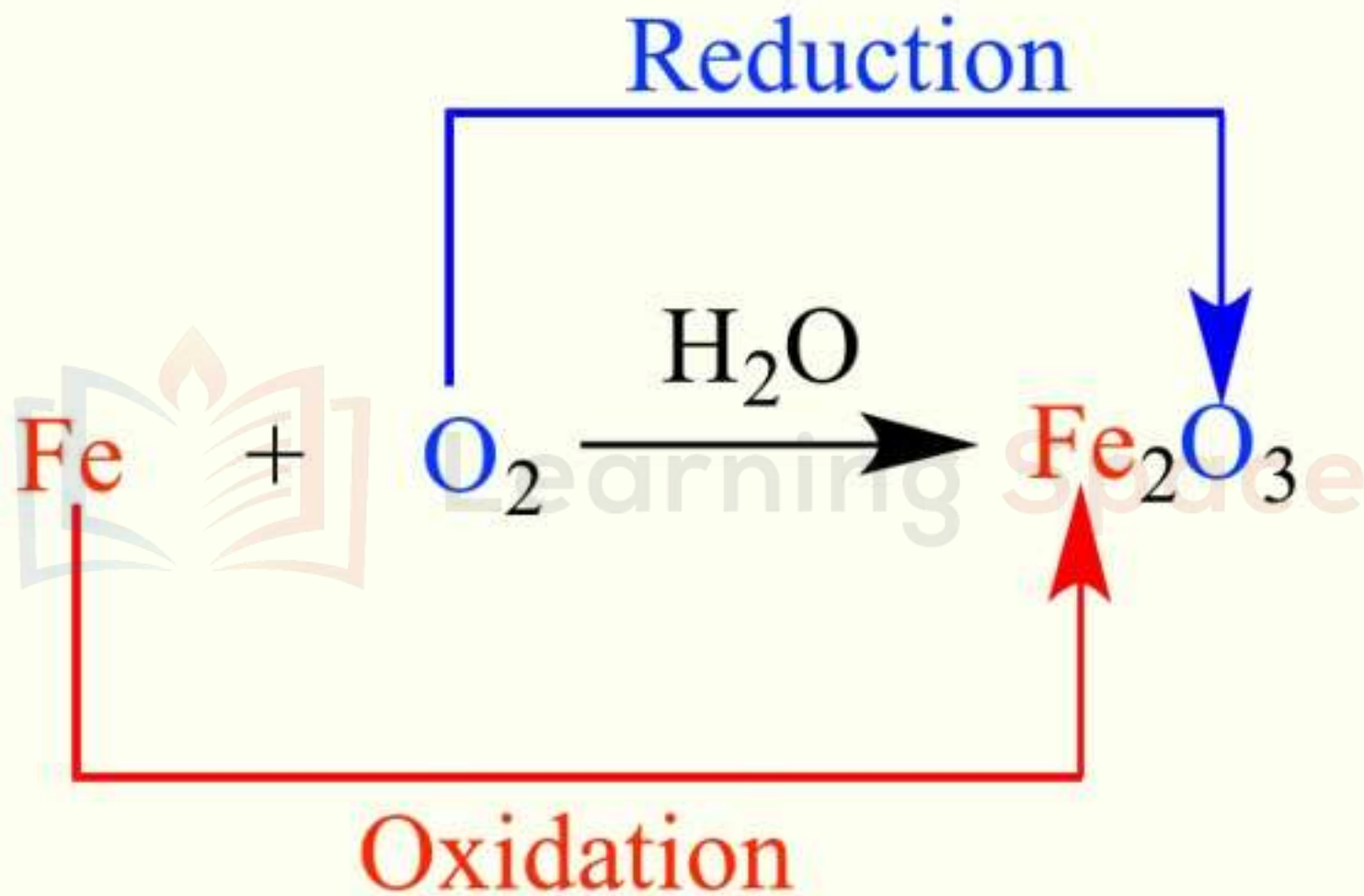
DOUBLE DISPLACEMENT REACTIONS

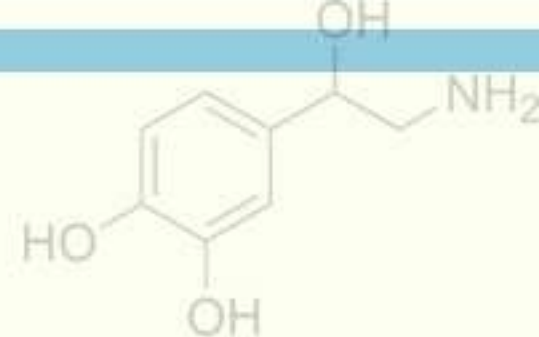


- Interchange of ions
- Results in formation of two new salts from two different salt solutions
- One of the product may be insoluble in water
- Precipitation of one product is common.



REDOX REACTIONS





REDOX REACTIONS

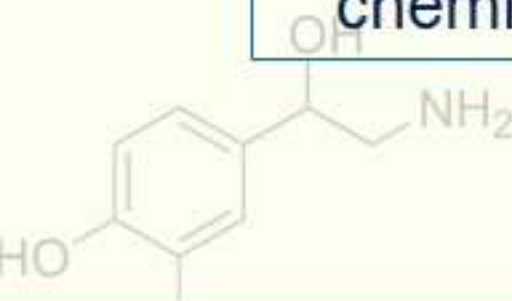
OXIDATION

- Results in Loss of electrons.
- Oxygen is added
- Hydrogen is removed
- Causes the increase of a positive charge in a chemical species

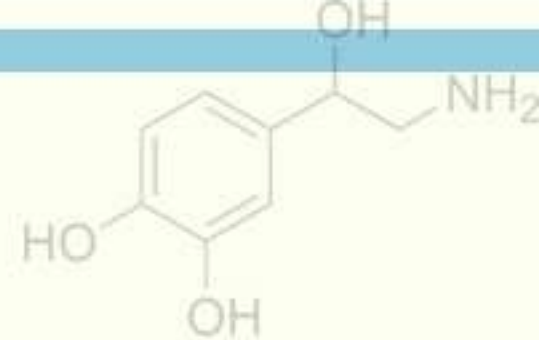
REDUCTION

- Results in gain of electrons.
- Oxygen is removed
- Hydrogen is added
- Causes the increase of a negative charge in a chemical species

***OIL** - oxidation is loss of electrons*
***RIG** - reduction is gain of electrons*

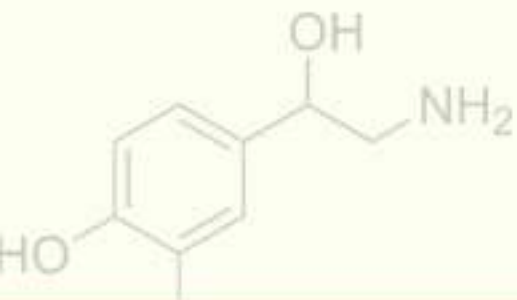
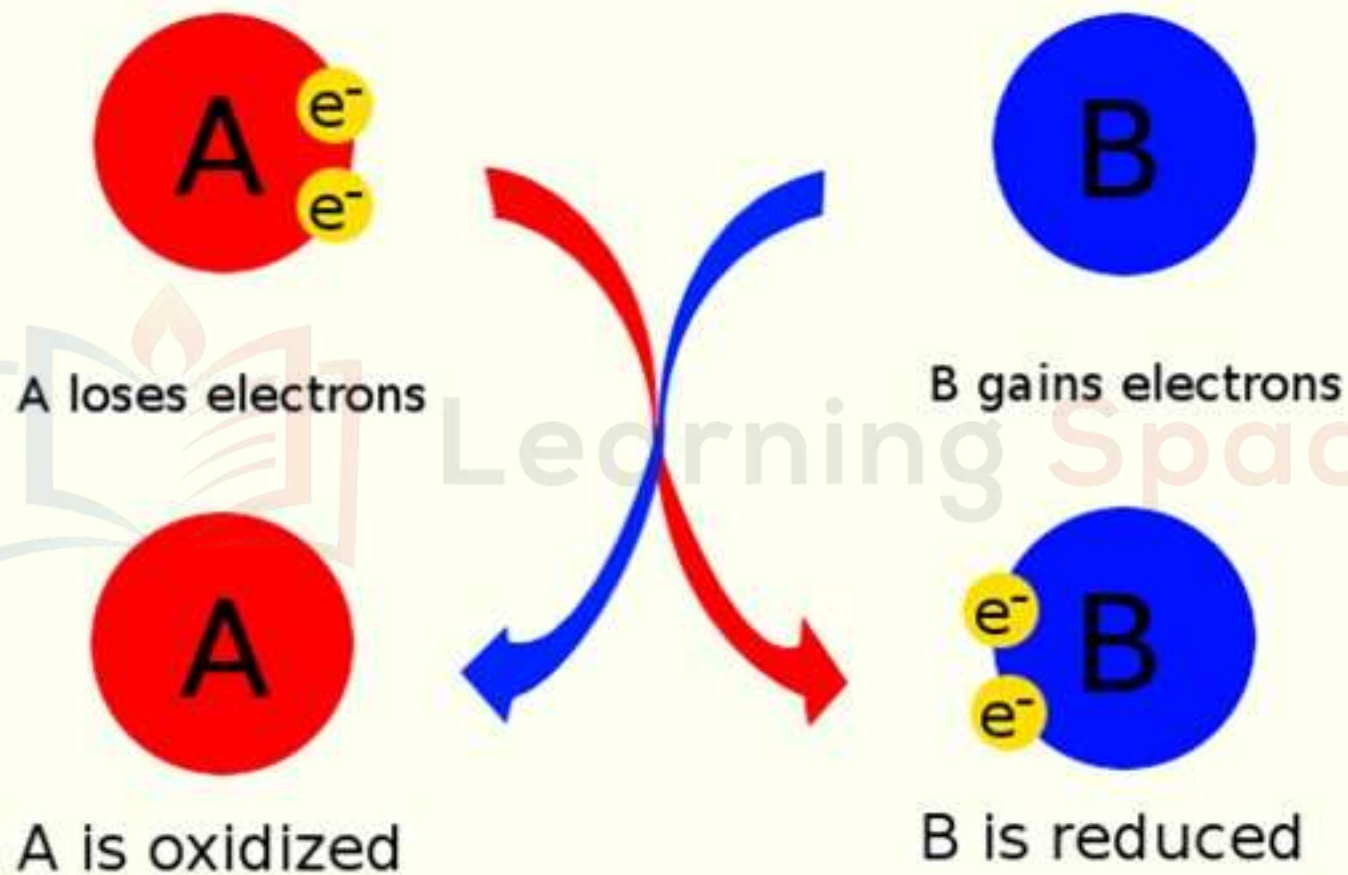


OXIDISING AND REDUCING AGENT



Reducing Agent

Oxidizing Agent



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